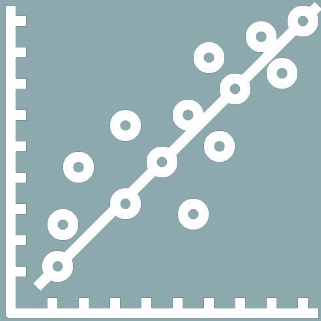




Lesson 6: Linear Regression

Business Analytics Course | Spring 2024





01. Supervised and Unsupervised Learning

02. Simple Linear Regression

03. Excel Follow-Along

04. Multiple Linear Regression

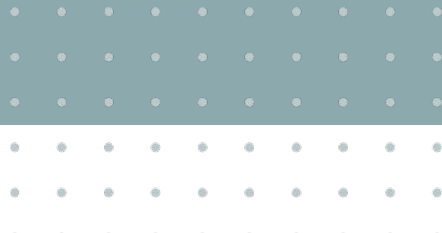
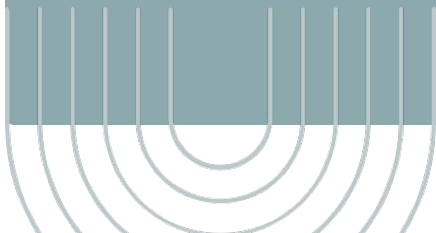


TABLE OF CONTENT



01.

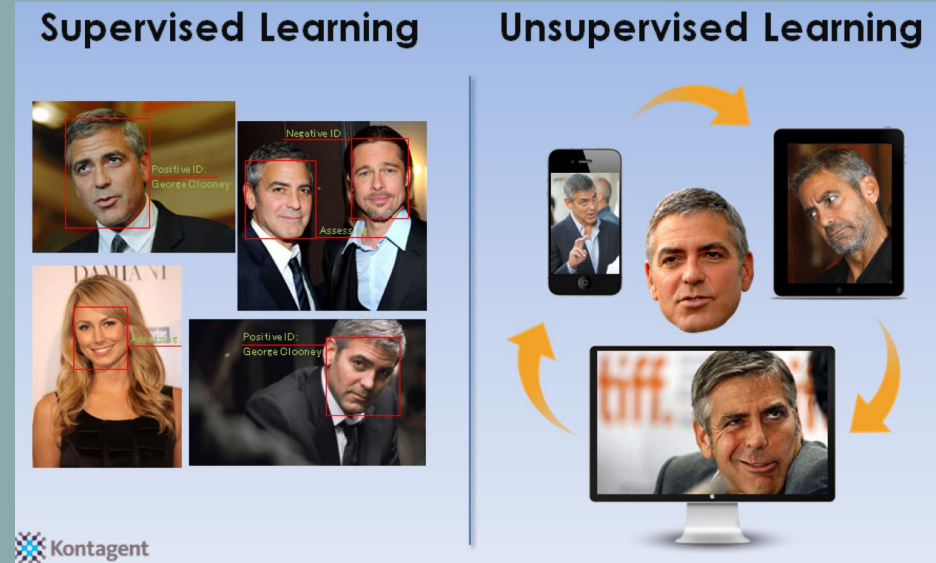
Supervised and Unsupervised Learning





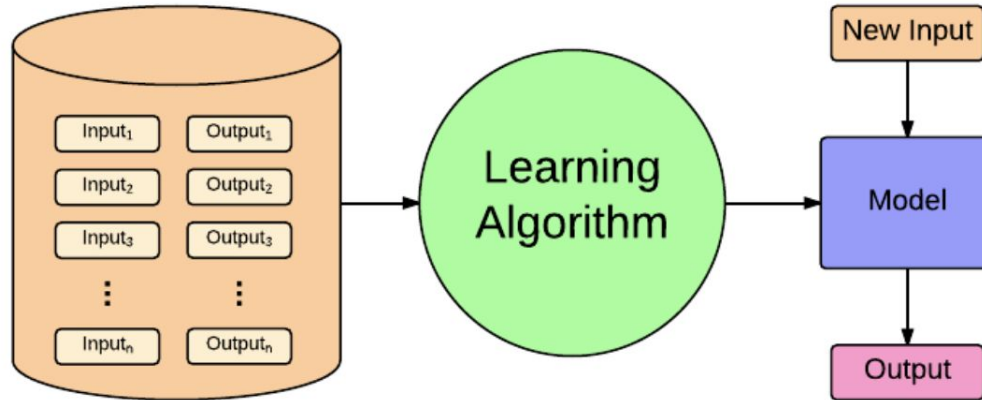
Supervised vs Unsupervised Learning

- Supervised learning algorithm uses a sample dataset to train itself to make predictions, iteratively adjusting itself to minimize error. These datasets are labeled for context, providing the desired output values to enable a model to give a “correct” answer.
- Unsupervised learning algorithms work independently to learn the data's inherent structure without any specific guidance or instruction. You simply provide unlabeled input data and let the algorithm identify any naturally occurring patterns in the dataset.



Supervised learning uses labeled training data, and unsupervised learning does not.

Supervised Learning Paradigm

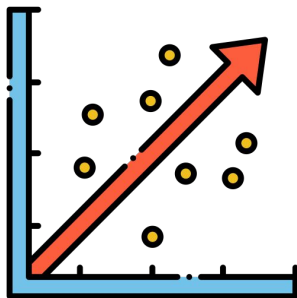


- Traditional statistics: domain experts work for many years to learn good features; they bring statisticians a small clean dataset
- Today's scenario: Domain knowledge is limited in new fields and large data sets are readily available. we (are sometimes forced to) start with a large dataset with many features

Supervised Learning Techniques



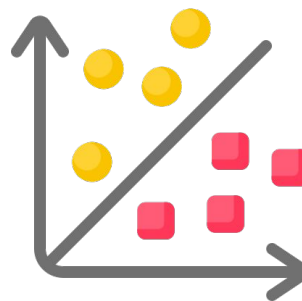
Regression



Use regression techniques when Y (dependent variable) is numerical

- Predicting home prices based on features
- Forecasting product demand based on historical sales data

Classification



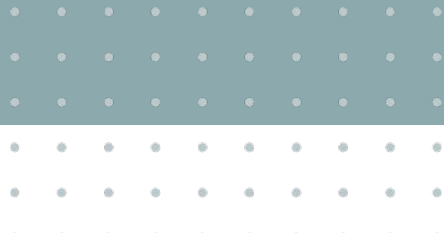
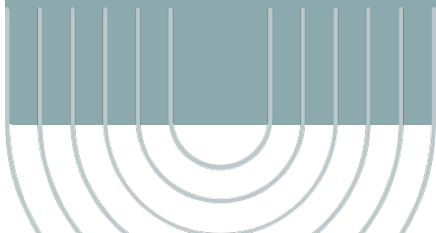
Use regression techniques when Y (dependent variable) is categorical

- If a transaction is fraudulent or not
- If the cell is benign or malignant
- If the email is spam or not



02.

Simple Linear Regression





What is Simple Linear Regression

Simple linear regression is used to estimate the relationship between **two quantitative variables**. You can use simple linear regression when you want to know:

1. How strong the relationship is between two variables (X and Y).
2. The value of the dependent variable (Y) at a certain value of the independent variable (X)

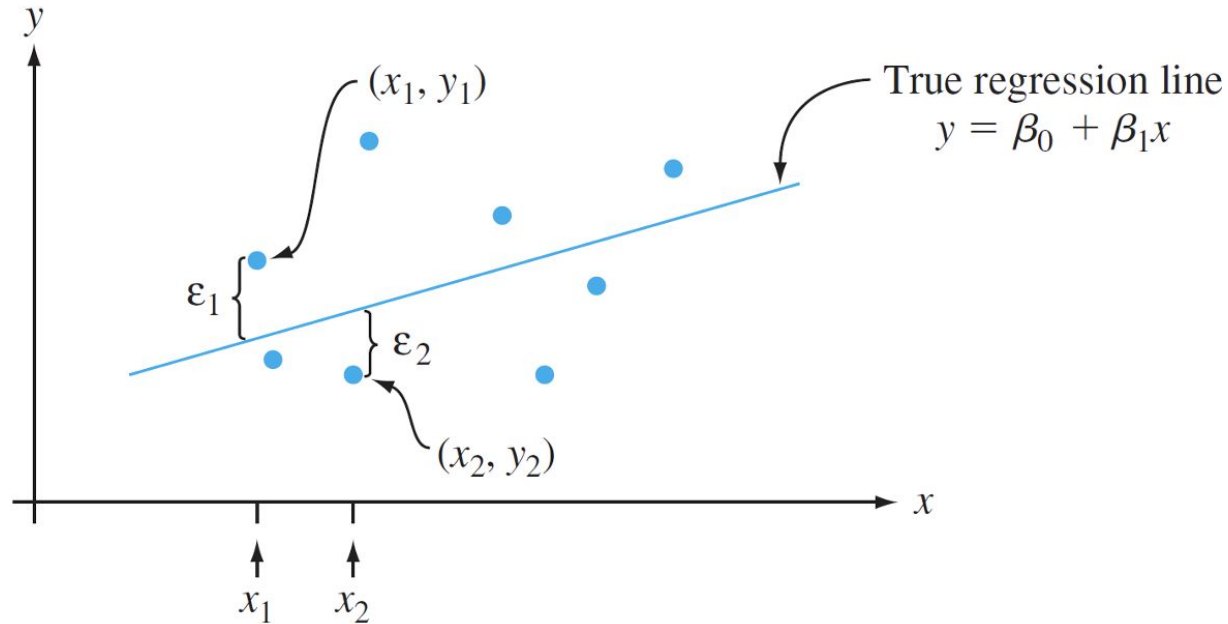
The formula for a simple linear regression is:

$$\textcircled{y} = \textcircled{\beta_0} + \textcircled{\beta_1} \textcircled{X} + \textcircled{\epsilon}$$

- y is the predicted value of the dependent variable for any given value of x
- β_0 is the intercept, the predicted value of y when the x is 0.
- β_1 is the regression coefficient – how much we expect y to change as x increases.
- X is the independent variable (the variable we expect is influencing y).
- ϵ is the error/variation of the estimate



Linear regression finds the line of best fit line through your data by searching for the regression coefficient (B_1) that minimizes the total error (e) of the model.



Assumptions of Linear Regression



Simple linear regression is a parametric test, meaning that it makes certain assumptions about the data. These assumptions are:

1. **Homogeneity of variance (homoscedasticity):** the size of the error in our prediction doesn't change significantly across the values of the independent variable.
2. **Independence** of observations: the observations in the dataset were collected using statistically valid sampling methods, and there are no hidden relationships among observations.
3. **Normality:** The data follows a normal distribution.

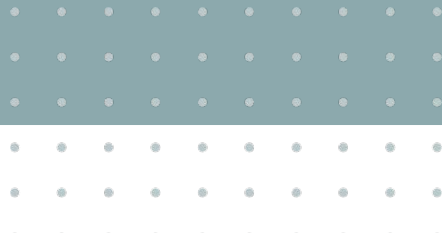
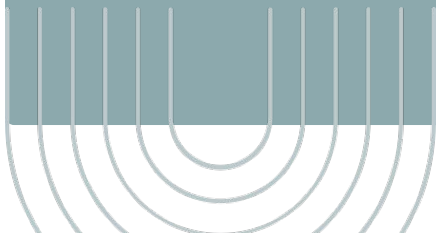
Linear regression makes one additional assumption:

4. The **relationship between the independent and dependent variable is linear:** the line of best fit through the data points is a straight line (rather than a curve or some sort of grouping factor).



03.

Excel Example Follow-Along



Example: Direct Marketing Data



- A direct marketing firm has a data set containing information on past customer behaviors
- It wants to determine which customers to target
- **Question: "Which customer characteristics relate to *AmountSpent*, and how?"**

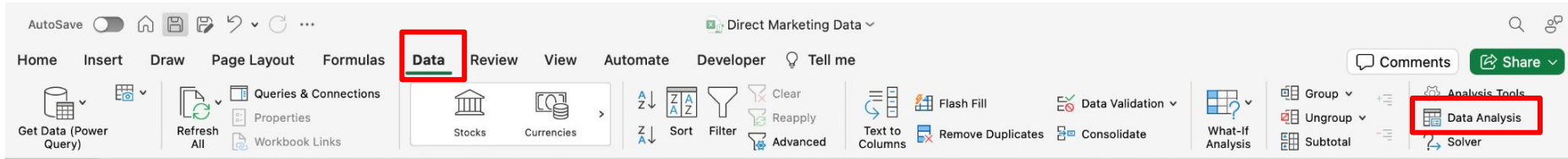
Age	Gender	OwnHome	Married	Location	Salary	Children	History	Catalogs	AmountSpent
Young	Male	Rent	Single	Close	15000	3	Low	6	38
Young	Male	Rent	Single	Close	13000	3	Low	6	43
Young	Female	Rent	Single	Close	14600	3	Low	6	47
Young	Female	Rent	Single	Close	17900	3	Low	6	62
Old	Female	Own	Single	Close	12700	2	Low	6	65
Young	Female	Rent	Single	Close	23000	3	Low	6	79
Young	Female	Rent	Single	Close	12700	3	Low	12	87
Young	Female	Rent	Single	Close	12100	1	Low	6	90
Old	Female	Own	Married	Close	10100	1	Low	6	93
Young	Female	Rent	Married	Close	42000	3	Low	6	105
Young	Male	Rent	Single	Close	11200	0	NA	6	106

Please download the sheet [here](#) (File → Download → Microsoft Excel) and follow along :)

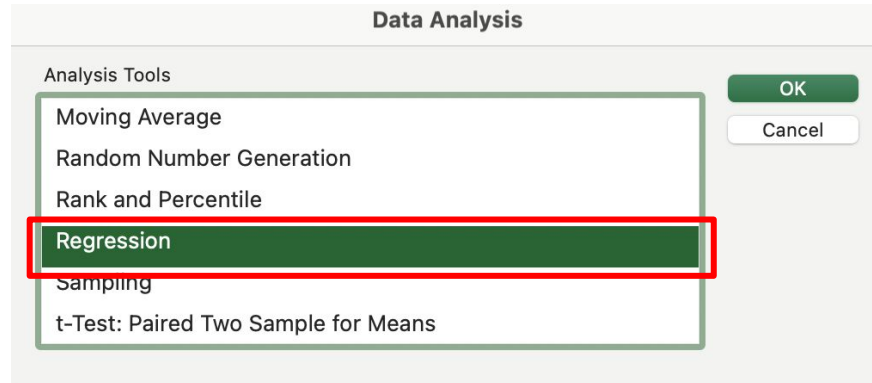
Linear Regression on Excel – 1:



STEP 1: Go to the DATA ANALYSIS Under DATA Tab



STEP 2: Scroll down and choose REGRESSION



Linear Regression on Excel – 2:



STEP 3: Input Y and Y Range and Choose desired output range

Regression

Input

Input Y Range:

Input X Range:

☒ Labels ☐ Constant is Zero

☐ Confidence Level: %

Output options

☒ Output Range:

☐ New Worksheet Ply:

☐ New Workbook

Residuals

☐ Residuals ☒ Residual Plots

☐ Standardized Residuals ☒ Line Fit Plots

Normal Probability

☐ Normal Probability Plots

OK
Cancel

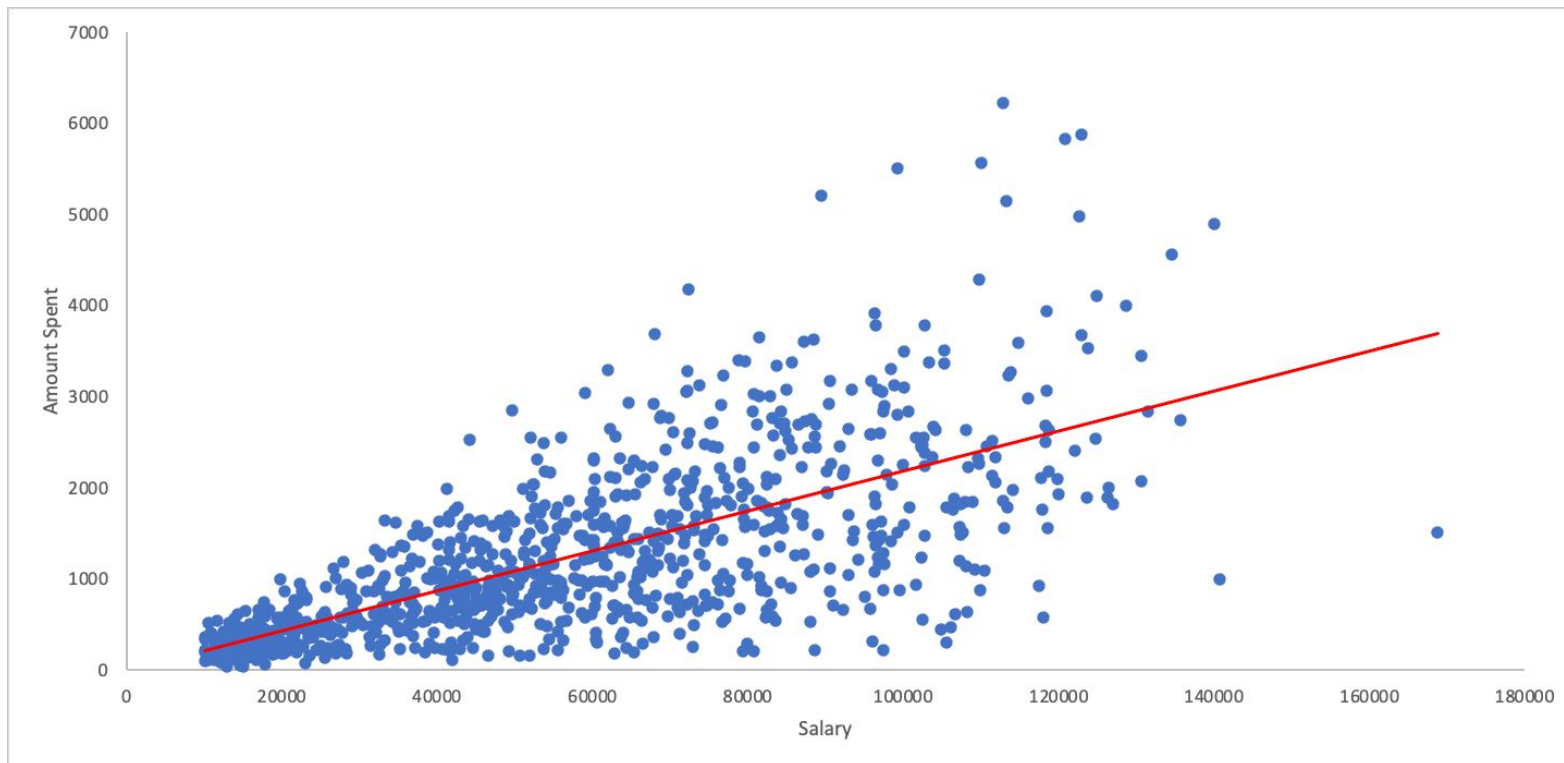
X & Y Range

Check Labels box if
you include the header
label in the range

Which cell do you
want your regression
model to be at?

Check the plots if
you want any plots
along the
regression model

Let's explore the relationship of AmountSpent (Y) & Salary (X)



What Do We Learn from the Model?



R Square: About 48.94% of the variability in Amount Spent can be explained by Salary

Intercept (β_0): The Average Amount Spent when Salary is \$0 is \$-15.31

Salary (β_1): If the salary of a customer increases by \$1, the Amount Spent is predicted to be increased by \$0.02

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.69959571
R Square	0.48943415
Adjusted R Square	0.48892256
Standard Error	687.064877
Observations	1000

ANOVA

	df	SS	MS	F	Significance F
Regression	1	451615196.5	451615197	956.6940035	7.4954E-148
Residual	998	471114028.6	472058.145		
Total	999	922729225.1			

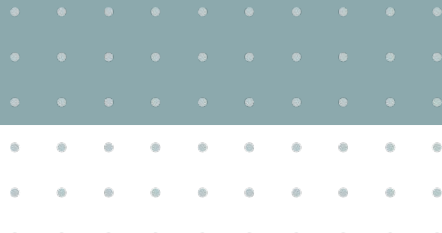
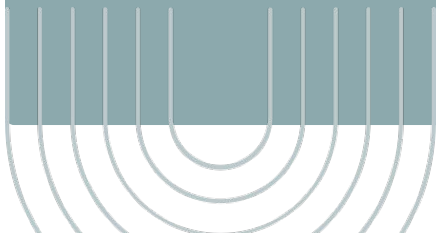
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-15.317825	45.37416253	-0.3375892	0.735743803	-104.357534	73.7218832	-104.35753	73.7218832
Salary	0.02196082	0.000710006	30.9304705	7.50E-148	0.020567547	0.0233541	0.02056755	0.0233541

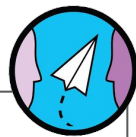
$$\text{AmountSpent} = -15.31 + 0.021 * \text{Salary} + \epsilon$$



04.

Multiple Linear Regression





What is Multiple Linear Regression

Multiple linear regression is used to estimate the relationship between **two or more independent variables and one dependent variable**. You can use multiple linear regression when you want to know:

1. How strong the relationship is between two or more independent variables and one dependent variable
2. The value of the dependent variable at a certain value of the independent variables

The formula for a multiple linear regression is:

$$y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \epsilon$$

- y = the predicted value of the dependent variable
- β_0 = the y-intercept (value of y when all other parameters are set to 0)
- $\beta_1 X_1$ = the regression coefficient (β_1) of the first independent variable (X_1) (a.k.a. the effect that increasing the value of the independent variable has on the predicted y value)
- $\dots$ = do the same for however many independent variables you are testing
- $\beta_n X_n$ = the regression coefficient of the last independent variable
- ϵ = model error (a.k.a. how much variation there is in our estimate of y)

What Do We Learn from the Model?



Intercept (β_0): The Average Amount Spent when Salary = \$0, Catalog = 0, and Children = 0 is \$-442.76

Catalog (β_1): Controlling for salary and children, if we send out 1 more Catalog, the AmountSpent will be increased by \$47.70

Salary (β_2): Controlling for catalog and children, if the salary of a customer increases by \$1, the Amount Spent is predicted to be increased by \$0.02

Regression Statistics	
Multiple R	0.81144
R Square	0.658435
Adjusted R Square	0.657407
Standard Error	562.5277
Observations	1000

R Square: About 65.84% of the variability in Amount Spent can be explained by Salary, Number of Catalog sent to customers, and Number of Children

- Note that this model is better compared to the previous one because it can capture more variability

ANOVA

	df	SS	MS	F	Significance F
Regression	3	607557577.7	2.03E+08	639.9976566	9.558E-232
Residual	996	315171647.4	316437.4		
Total	999	922729225.1			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-442.7579	53.71765795	-8.242317	5.28715E-16	-548.170719	-337.3452	-548.1707	-337.3452
Catalogs	47.69532	2.75528387	17.31049	6.80682E-59	42.28849708	53.10215	42.2885	53.10215
Salary	0.020406	0.000592904	34.41674	1.1637E-171	0.019242335	0.021569	0.019242	0.021569
Children	-198.6947	17.08742942	-11.62812	2.13219E-29	-232.226184	-165.1632	-232.2262	-165.1632

Children (β_3): Controlling for catalog and salary, if the customer has 1 more child, the Amount Spent is predicted to be decreased by \$198.69

$$\text{AmountSpent} = -442.76 + (47.7 * \text{Catalog}) + (0.02 * \text{Salary}) + (-198.69 * \text{Children}) + \epsilon$$

Assignment



Download a diamond dataset from here (File → Download → Microsoft Excel): [Diamond Dataset](#)

carat	cut	color	clarity	depth	table	price	x	y	z
0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
0.29	Premium	I	VS2	62.4	58	334	4.2	4.23	2.63
0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75

Given a dataset of diamonds detailing carat, price, cut, color, and other attributes:

1. Build a simple linear regression model with Price as the dependent variable (Y) and Carat as the only independent (X)
2. Find the R-squared value and interpret the result
3. Interpret the slope coefficient B1

Assignment Answer



R Square: About 90.71% of the variability in Price can be explained by Carat

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9524123
R Square	0.9070892
Adjusted R Square	0.90614113
Standard Error	216.296564
Observations	100

ANOVA

	df	SS	MS	F	Significance F
Regression	1	44761959.45	44761959.5	956.775068	2.29718E-52
Residual	98	4584851.939	46784.2035		
Total	99	49346811.39			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-731.064	50.08355145	-14.596888	2.5439E-26	-830.453181	-631.67483	-830.453181	-631.6748253
carat	4329.88535	139.9817673	30.9317809	2.2972E-52	4052.096091	4607.67461	4052.096091	4607.674613

Carat (β_1): If the weight of the diamond increases by 1 carat, the Price is predicted to be increased by \$4329.86



THANK YOU

Have any question?